SAL EDUCATION

SAL ENGINEERING AND TECHNICAL INSTITUTE

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Student's Weekly Report of Internship

Student Name:	**Thakkar Krinal Balkrushanbhai**			
Enrollment Number:	**231263116020**			
Name Of Organization:	**Demaze Technologies**			
External Guide Name:	**Deep Suthar**			
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Week 3: React Basics

Monday:

Introduction to React & JSX

React is a JavaScript library used for building fast and interactive user interfaces. It allows developers to create reusable components and manage application state efficiently. JSX (JavaScript XML) is a syntax extension used in React that allows HTML-like code to be written inside JavaScript, making UI development easier and more readable. React improves performance by updating only the required parts of the user interface using a virtual DOM.

```
script.js > default
import React from 'react';

function App() {
    return <h1>Hello Krinal</h1>;
}

export default App;
```

Tuesday:

Components & Props

Components are reusable building blocks in React used to create user interfaces. Props (properties) are used to pass data from one component to another, allowing components to display dynamic content. Props make components reusable and flexible.

- Components are reusable UI building blocks
- Two types: Functional and Class components
- Props are used to pass data to components
- Props are read-only (cannot be changed)
- Help in creating dynamic and reusable UI

```
function Student(props){
    return <h2>Hello {props.name}</h2>;
}
```

Wednesday:

State and setState in Class components

In React class components, state is an object that holds data which can change over time. It is used to manage dynamic behavior in a component. The setState() method is used to update the state in a controlled way. Whenever the state is updated using setState(), React automatically re-renders the component to reflect the changes in the user interface.

Main Points:

- State stores dynamic data of a component
- State is local and private to the component
- setState() is used to update state
- Direct modification of state is not allowed
- Updating state re-renders the component

```
Welcome    myapp.js ×
myapp.js > App
1 import React, { Component } from 'react';
2
3 class App extends Component {
4     state = { message: "Hello" };
5     changeMessage = () => { this.setState({ message: "Welcome
      Krinal" }); };
6     render() { return <div><h2>{this.state.message}</h2></
      div>; }
7 }
```

Thursday:

useState hook in Functional components

The useState hook allows functional components to manage state in React. It provides a simple way to store and update data within a component. When the state is updated using the provided function, React automatically re-renders the component, making the user interface dynamic and responsive.

Main Points:

- useState is a React hook used to manage state
- It is used in functional components
- It returns state value and update function
- State changes re-render the component
- Makes functional components stateful

```
Welcome    myapp.js ×
myapp.js > ...
1 import React, { useState } from 'react';
2 function App() {
3     const [msg, setMsg] = useState("Hello");
4     return <button onClick={() => setMsg("Welcome")}>{msg}</
      button>;
5 }
6
```

Friday:

- Small project: Counter App using React

Summary of Work Done Till Date

- Learned introduction to React and JSX
- Studied components and props
- Understood state management in class components
- Learned useState hook in functional components
- Developed a Counter App using React

Plans for next week:

- Learn intermediate React concepts
- Practice conditional rendering
- Understand list rendering and keys
- Build small interactive React projects

Total Working Hrs: _____ **Signature of Student:**

The above entries are correct and the grading of work done by Trainee is

Excellent	Very Good	Good	Fair	Below Average	Poor

Signature of External Guide with Company Seal: **Signature of Internal Guide**

Date: **Date:**